Jheronimus Academy of Data Science

Reflection Report: format for the reflection report

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A brief description of the societal or business impact of your thesis

(You should describe how your solution could lead to a revenue increase, a cost decrease, or both)

The aim of my thesis was to predict the contract award price for a public procurement contract. In doing so, contracting authorities can reduce costs through more accurately estimating award prices and through assist in establishing/guiding tender budget development, or use it to evaluate the tender, saving costs through reducing required manual labour. Furthermore, the algorithm can be used to adjust the contract to assess the impact on contract award price and adjust the content of the contract accordingly.

A brief description of the ethical considerations and screening steps taken to adhere to the Ethical Review Board requirements[1]

Ethical considerations which should be taken into account in the context of this thesis include: Implementation of algorithms upon which contracting agencies become too reliant, possibly resulting in situations where too high or too low price are demanded from the market, disrupting regular market mechanics as envisioned by the European Union Directives guiding public procurement and the fundamental pillars upon which it is build such as open and fair competition.

However, in line with the guidelines as set out by the IRB, no research "with data relating to living humans or groups of humans" is conducted for which exemptions are required.

Links to the code and data used for the master thesis as published on an open platform

(In case your data and code are confidential, please provide the documents indicating both supervisors in your portfolio have checked the data and code. This document can be found on Canvas)

All data is extracted from the TED database, a website collecting, storing and disseminating information on public procurement to in line with the EU directives. Link: https://ted.europa.eu/TED/main/HomePage.do

[1] https://www.tilburguniversity.edu/research/economics-and-management/institutional-review-board

In order to download the data from the TED website, a free account has to be made.

Division of tasks and responsibilities

(This part only has to be filled in, in case you did your graduation project in a team of 2-3 students)

Reflection on the overall learning goals for the master thesis

1. **Design, Implement, and Market Novel Data-Centered Products and Services**

No studies exist exploiting large language models, and no combination has been made between structured and unstructured data in public procurement. Currently, only statistical approaches to incorporating text features in modelling the relationship between contract information and competition, collusion, and cost effectiveness (Gorgun et al., 2019), or using relatively simplistic NLP methods like n-grams which have been around for a long time (Gorgun et al., 2020), with a first attempt to predict the contract award price. This required a noval approach to data collection, cleaning and modelling to bring about a prediction algorithm that was not explored in previous literature. In this sense, the focus is on implementing novel data-centered products and services in a new realm.

On a practical level, it required an in-depth understanding of complicated transformer models and how to combine them with other neural network models to create combined neural networks. Aspects of the work included working with constraint resources, managing GPU, requiring clever data cleaning practices as the data quality was really poor, requiring an effective and efficient design to facilitate a feasible workflow.

2. **Detect and Create New Business Opportunities**

The integration of structured and unstructured data in predicting contract award prices opens doors to novel business opportunities in the field of public procurement. The potential for contracting authorities to reduce costs through more accurate estimations of award prices and informed establishment of tender budgets becomes apparent. Furthermore, the algorithm lends itself well for evaluating the content of a tender by predicting the contract award price, allowing contracting agencies to make adjustments accordingly. An example of such an adjustment includes the use of adequate information in the content sections which are analyzed using the transformer component, saving up to 6-8% on final award price (Gorgun et al., 2022). In addition, more accurate award prices may install a sense of trust among suppliers as to their evaluation of the contract's value, resulting in a larger number of bids, and also signal a greater level of competence (Rhode, 2018). In essence, the developed methodology

becomes a valuable asset for contracting authorities, streamlining decision-making processes and optimizing resource allocation.

3. **Critically Evaluate and Advise on Economic and Social Sustainability**

Within the exploration of unstructured data and open data source like TED lies opportunities for future research and entrepreneurial exploitation. While complexities may have limited the current study from delving into topics such as topic modeling or employing language models to understand very specific details of the contract such contract award criteria and the weights assigned to it, recognizing the potential value in these areas is crucial. The interplay between unstructured, structured, open data source and economic and social sustainability in public procurement processes presents a rich landscape for future investigations.

4. **Formulate a Research Question and Problem Statement**

During my thesis, I formulated a research question and problem statement based on a perceived business problem in data science and entrepreneurship field. The question centered on predicting contract award prices in public procurement, addressing a significant challenge in the field with only a few other studies.

5. **Motivate the Problem Statement and Specify the Research Approach**

I motivated the problem statement, highlighting the importance of predicting contract award prices for contracting authorities. The research approach was specified, outlining the integration of structured and unstructured data and the deployment of advanced machine learning models.

6. **Indicate the Scientific and Societal Relevance of the Research Question**

The scientific and societal relevance of the research question was clearly indicated throughout the thesis. The potential impact on cost reduction, informed decision-making, and resource optimization in public procurement highlighted the societal significance, while the exploration of novel methodologies contributed to the scientific relevance. For a more elaborate answer see the literature review of the thesis.

7. **Select and Review Appropriate Literature and Describe Relevan Theories**

The literature review in my thesis was comprehensive and well-structured. I selected and reviewed appropriate literature, providing insights into existing studies on public procurement, contract award prices, and the use of structured and unstructured data. Relevant theories were presented in an understandable and convincing manner.

8. **Position the Problem at Hand Concerning the Theory and Present the Conceptual Framework**

I have not used a particular framework for working on my thesis other than the standard scientific approach. I considered applying the CRISP-DM framework but it held little relevance in the end for guiding the process

9. **Describe Methods Used in the Research in a Clear Manner**

The methods used in my research were described in a clear manner covering standard topics like data imputation, train test splits, encoding etc. The data collection process from the TED database was outlined, and the combination of multilingual transformer models with other neural network models

10. **Present Results in a Structured Manner and Formulate Recommendations**

I am still have to do my final presentation but the thesis obviously required a lot of attention to format properly, positioning, labeling and designing every table for the purpose of conveying a proper representation of the data.

11. **Formulate Clear Recommendations and Specific Actions and Initiatives**

Clear recommendations and specific actions and initiatives were formulated based on the research findings. These recommendations addressed the potential for cost reduction, informed budget development, and adjustments to tender content, showcasing the practical applications of the research in the field of public procurement. Furthermore, a series of future research directions were formulated including the comparison of various language models, additional text features, or the use of transformer models for other prediction tasks than the contract award price.